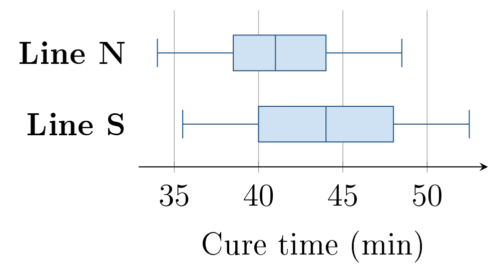
Two Test Plans for Cure Time

Unit 4 · Topic 4.9 · TODAY'S PROBLEM

Suppose line N uses a new formula for sealant, a material that closes gaps. Line S uses the standard formula. A team takes independent simple random samples of 18 of 600 recent N batches and 18 of 540 recent S batches. Mean cure times are 41.2 and 44.0 minutes; the boxplots are shown. Before sampling, the team asked, "Do the two populations differ in mean cure time?" Formula was not randomly assigned to lines.



Rhea: "Keep the original question and use population means. A difference between lines may have other causes."

Sol: "Since $41.2 < 44.0$, use a lower tail with sample means. A significant result proves the formula caused faster curing."

FIRST TAKE (5 min, on your sheet)

Did the team originally ask whether N was faster or whether the lines differed in either direction? Explain in one sentence.

DOK 1 **(a)** Define μ_N and μ_S in context.

DOK 2 **(b)** State H_0 and H_a in N-minus-S order to match the question asked before sampling.

DOK 3 **(c)** In 3–4 sentences, judge the two plans. Explain why the original question sets the tail, why hypotheses use population means, and why these two lines cannot isolate the formula as the cause.

Optional review: follow-along (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.